
CS 581 Online

Introduction to Electronic Health Records

Live Classroom

HIS/EHR Clinical Functionality

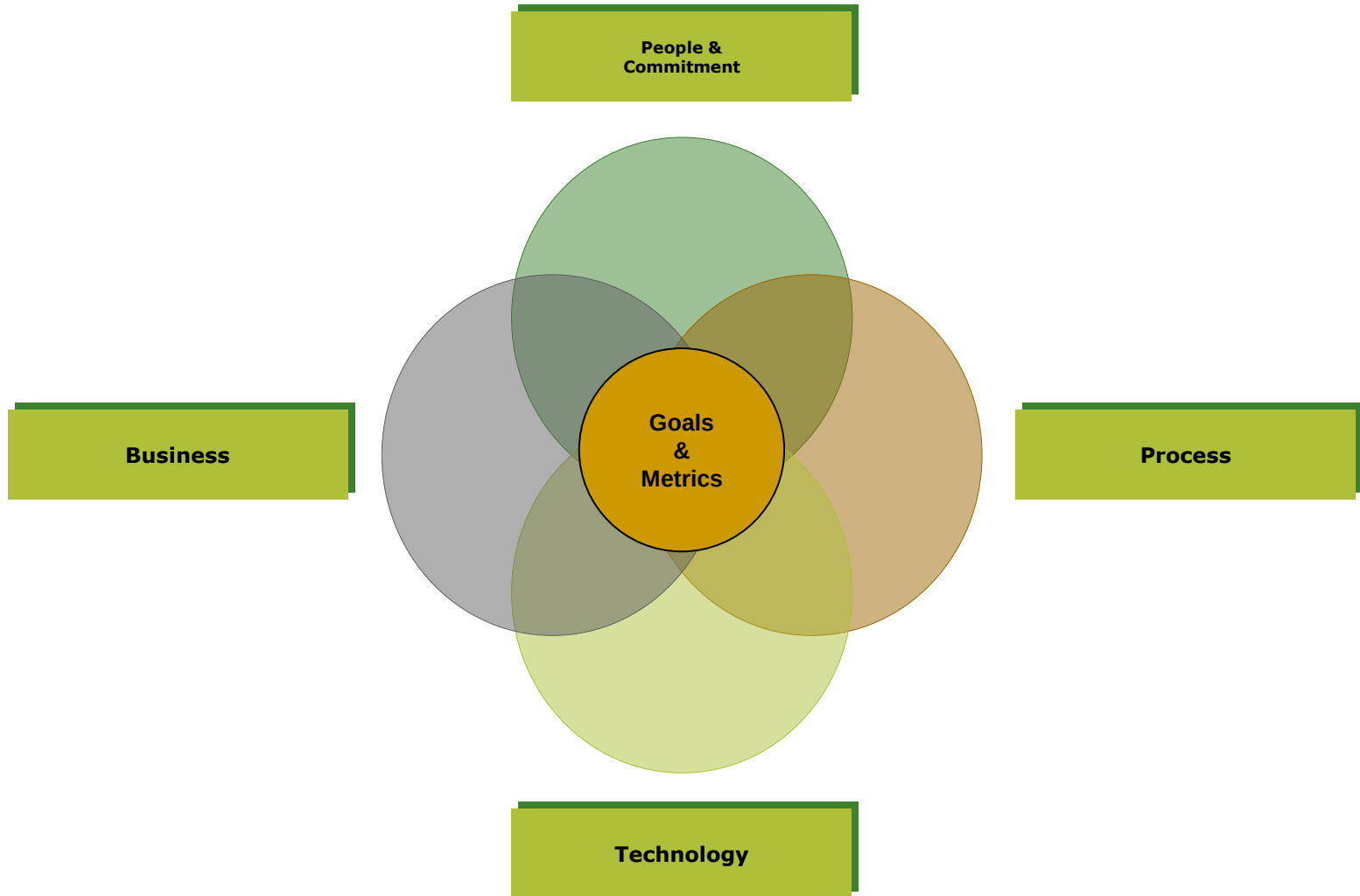
Michael Levinger

January 22, 2025

Agenda

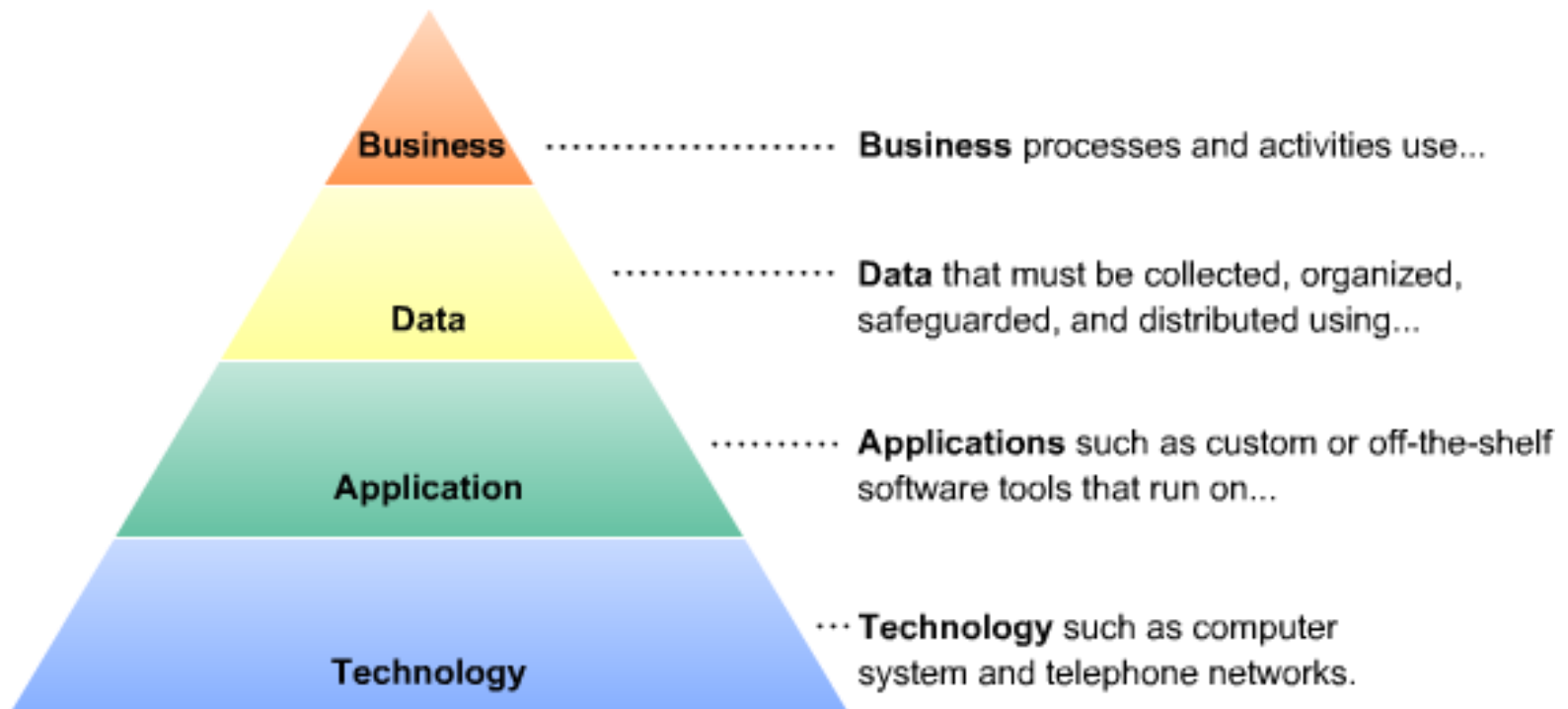
- Enterprise Architectures
- HIS/EHR Functionality Overall
- Clinical Functionality - EHR Functionality

HIS/EHR Balanced Performance Model



Enterprise Architecture Overview

TOGAF

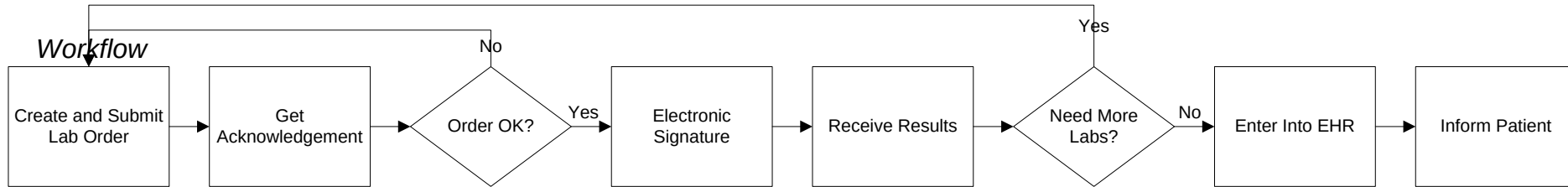


Enterprise Architecture applied to an EHR

- **Business**—The processes and activities required to operate the business. This is part of BPR and the movement from As-Is to Should-Be processes. For example, with an EHR the process a patient goes through visiting a physician.
- **Data**—All of the information that the business processes require and which must be stored, organized, protected, and distributed securely. In the case of a patient visiting a physician the information about the patient – name, symptoms, diagnosis, treatment, etc.
- **Application**—The application system(s) to which use or create the data. In our case the HIS/EHR application.
- **Technology**—The underlying computer system and networks required to run the application. Like any application system an HIS/EHR requires servers, networks, etc. to operate.

Diagnostic Test Ordering – Example BPR

Workflow



Data Needed

•Diagnosis •Possible labs	•Lab acknowledge ment	•Original Lab Order •Lab Ack. •Diagnosis	•Physician ID	•Original Lab Order •Lab Results •Diagnosis	•Original Lab Order •Lab Results •Diagnosis	•Lab Results •Diagnosis	•Patient Info •Lab Results •Diagnosis
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Who is involved?

•Physician •Nurse •Lab	•Physician •Nurse •Lab	•Physician •Nurse	•Physician	•Physician •Nurse •Lab	•Physician	•Physician •Nurse •Admin Staff	•Physician •Nurse •Patient
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EHR Functionality

•Order Sets •Order Protocols •Lab System (LIMS) Interface	•Lab System (LIMS) Interface	•Clinical Decision Support •Order Protocols	•Electronic Signature	•Lab System (LIMS) Interface	•Clinical Decision Support •Order Protocols	•Lab System (LIMS) Interface	•Patient Communication
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Benefit

•Faster •Less Error Prone	•Faster •More available	•Faster •Less Error Prone	•Safer •Faster	•Faster •Greater availability	•Faster •Less Error Prone	•Faster •Greater availability	•Faster •Greater availability •Improved Communication
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Agenda

- Enterprise Architectures
- HIS/EHR Functionality Overall
- Clinical Functionality - EHR Functionality

What functions are parts of a Health Information System?

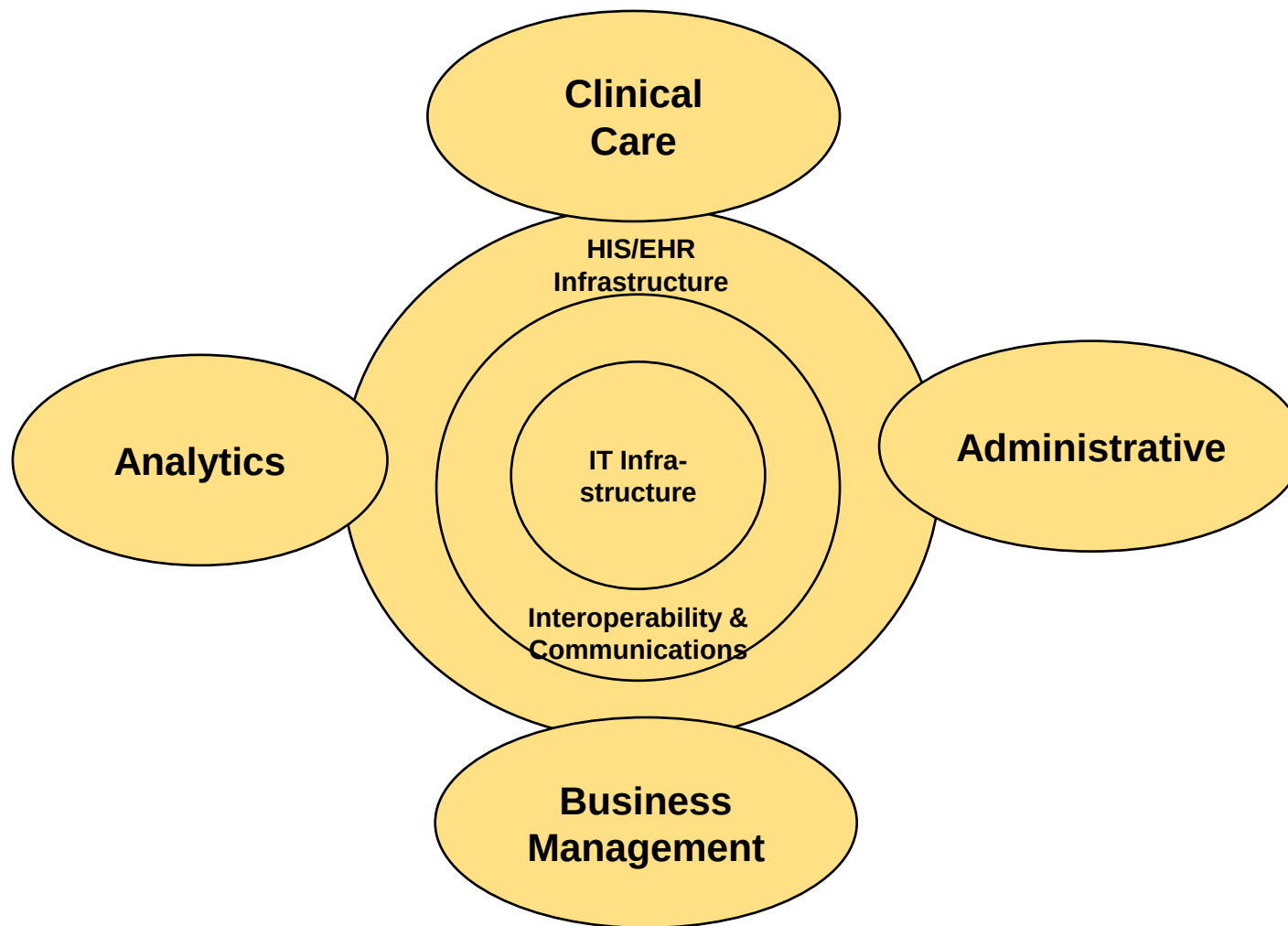
Small Group Discussions



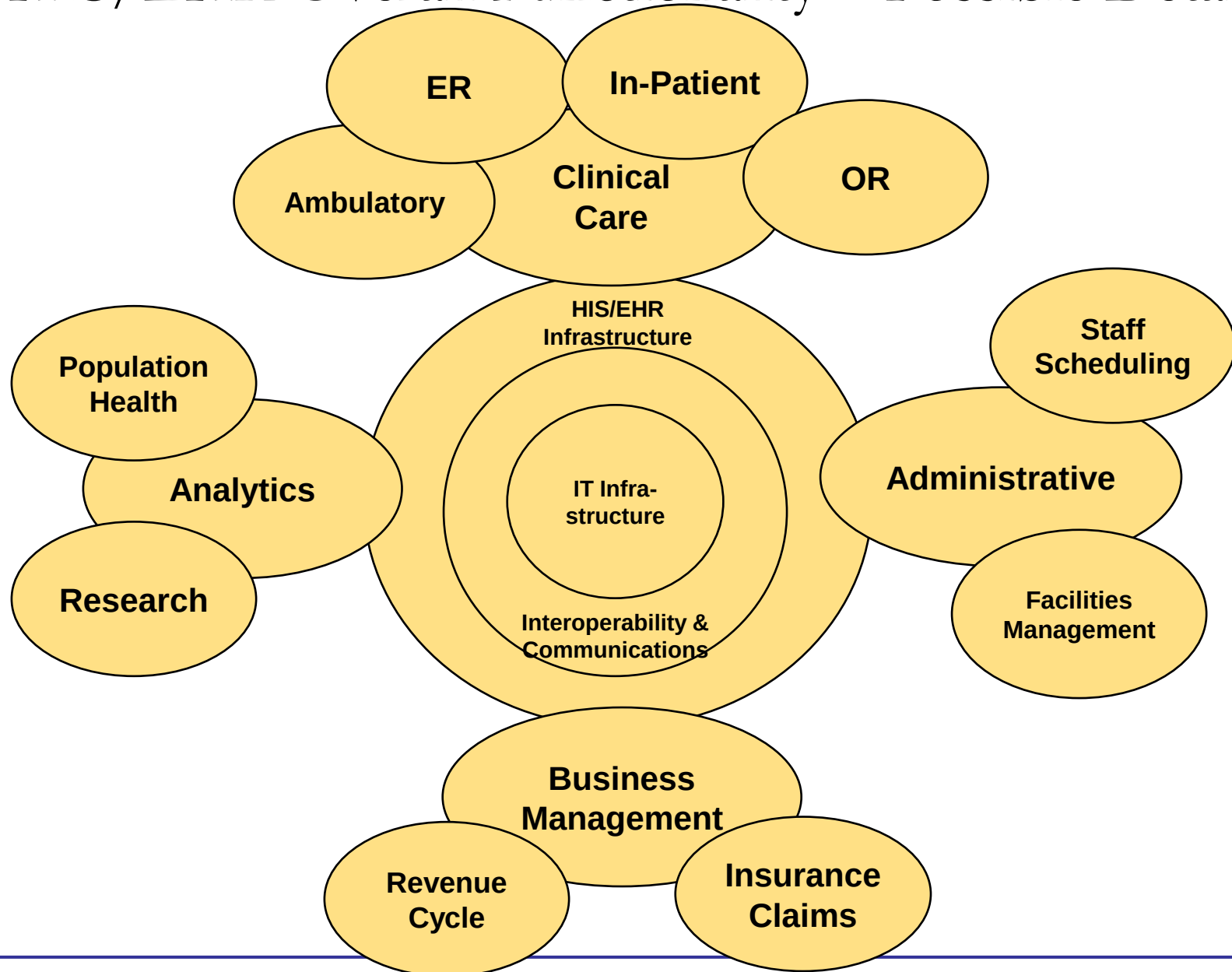
Instructions

- You will be assigned to a Zoom breakout group for discussion
- The groups will have approximately 10 minutes to discuss the topic and pick a spokesman
- Then we will reconvene and our group thoughts

HIS/EHR Overall Functionality

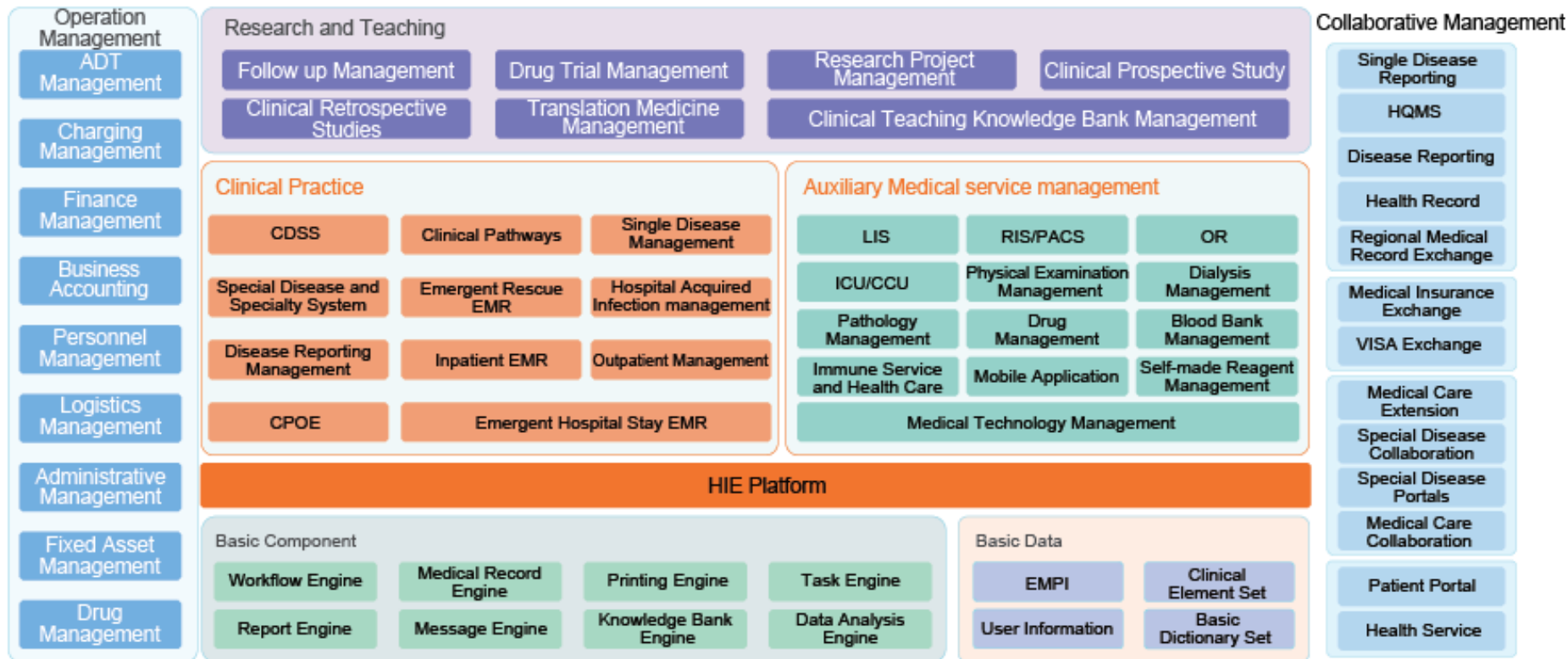


HIS/EHR Overall Functionality – Possible Detail



Application & Technology Layer

Integrated Hospital/Healthcare Information System



Source: <http://www.haitaiint.com/do?module=web&action=News&method=showIntSolutionTemp&solutionID=14082217291300000001>



What functions should be in an Ambulatory EHR?

What Functions are in an Ambulatory EHR?

Functions To Manage:

Manage Patient Records & Clinical Activities

Practice Operations and Financial Information

PATIENT MEDICAL HISTORY

PATIENT'S NAME: _____ DATE OF BIRTH: _____

ALTHOUGH DENTAL PERSONNEL PRIMARILY TREAT THE AREA IN AND AROUND YOUR MOUTH, YOUR MOUTH IS A PART OF YOUR ENTIRE BODY. HEALTH PROBLEMS THAT YOU MAY HAVE, OR MEDICATION THAT YOU MAY BE TAKING, COULD HAVE AN INTERACTIONS WITH THE DENTISTRY THAT YOU WILL BE RECEIVING. THANK YOU FOR ANSWERING THE FOLLOWING QUESTIONS.

Typical Clinical Functions

- 1. Allergies: ☐ YES ☐ NO
- 2. Current Medications: ☐ YES ☐ NO
- 3. Past Medical History: ☐ YES ☐ NO
- 4. Current Medications: ☐ YES ☐ NO
- 5. Allergies: ☐ YES ☐ NO
- 6. Current Medications: ☐ YES ☐ NO
- 7. Allergies: ☐ YES ☐ NO
- 8. Current Medications: ☐ YES ☐ NO
- 9. Allergies: ☐ YES ☐ NO
- 10. Current Medications: ☐ YES ☐ NO
- 11. Allergies: ☐ YES ☐ NO
- 12. Current Medications: ☐ YES ☐ NO
- 13. Allergies: ☐ YES ☐ NO
- 14. Current Medications: ☐ YES ☐ NO
- 15. Allergies: ☐ YES ☐ NO
- 16. Current Medications: ☐ YES ☐ NO
- 17. Allergies: ☐ YES ☐ NO
- 18. Current Medications: ☐ YES ☐ NO
- 19. Allergies: ☐ YES ☐ NO
- 20. Current Medications: ☐ YES ☐ NO
- 21. Allergies: ☐ YES ☐ NO
- 22. Current Medications: ☐ YES ☐ NO
- 23. Allergies: ☐ YES ☐ NO
- 24. Current Medications: ☐ YES ☐ NO
- 25. Allergies: ☐ YES ☐ NO
- 26. Current Medications: ☐ YES ☐ NO
- 27. Allergies: ☐ YES ☐ NO
- 28. Current Medications: ☐ YES ☐ NO
- 29. Allergies: ☐ YES ☐ NO
- 30. Current Medications: ☐ YES ☐ NO
- 31. Allergies: ☐ YES ☐ NO
- 32. Current Medications: ☐ YES ☐ NO
- 33. Allergies: ☐ YES ☐ NO
- 34. Current Medications: ☐ YES ☐ NO
- 35. Allergies: ☐ YES ☐ NO
- 36. Current Medications: ☐ YES ☐ NO
- 37. Allergies: ☐ YES ☐ NO
- 38. Current Medications: ☐ YES ☐ NO
- 39. Allergies: ☐ YES ☐ NO
- 40. Current Medications: ☐ YES ☐ NO
- 41. Allergies: ☐ YES ☐ NO
- 42. Current Medications: ☐ YES ☐ NO
- 43. Allergies: ☐ YES ☐ NO
- 44. Current Medications: ☐ YES ☐ NO
- 45. Allergies: ☐ YES ☐ NO
- 46. Current Medications: ☐ YES ☐ NO
- 47. Allergies: ☐ YES ☐ NO
- 48. Current Medications: ☐ YES ☐ NO
- 49. Allergies: ☐ YES ☐ NO
- 50. Current Medications: ☐ YES ☐ NO
- 51. Allergies: ☐ YES ☐ NO
- 52. Current Medications: ☐ YES ☐ NO
- 53. Allergies: ☐ YES ☐ NO
- 54. Current Medications: ☐ YES ☐ NO
- 55. Allergies: ☐ YES ☐ NO
- 56. Current Medications: ☐ YES ☐ NO
- 57. Allergies: ☐ YES ☐ NO
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- 60. Current Medications: ☐ YES ☐ NO
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- 65. Allergies: ☐ YES ☐ NO
- 66. Current Medications: ☐ YES ☐ NO
- 67. Allergies: ☐ YES ☐ NO
- 68. Current Medications: ☐ YES ☐ NO
- 69. Allergies: ☐ YES ☐ NO
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- 82. Current Medications: ☐ YES ☐ NO
- 83. Allergies: ☐ YES ☐ NO
- 84. Current Medications: ☐ YES ☐ NO
- 85. Allergies: ☐ YES ☐ NO
- 86. Current Medications: ☐ YES ☐ NO
- 87. Allergies: ☐ YES ☐ NO
- 88. Current Medications: ☐ YES ☐ NO
- 89. Allergies: ☐ YES ☐ NO
- 90. Current Medications: ☐ YES ☐ NO
- 91. Allergies: ☐ YES ☐ NO
- 92. Current Medications: ☐ YES ☐ NO
- 93. Allergies: ☐ YES ☐ NO
- 94. Current Medications: ☐ YES ☐ NO
- 95. Allergies: ☐ YES ☐ NO
- 96. Current Medications: ☐ YES ☐ NO
- 97. Allergies: ☐ YES ☐ NO
- 98. Current Medications: ☐ YES ☐ NO
- 99. Allergies: ☐ YES ☐ NO
- 100. Current Medications: ☐ YES ☐ NO

HEALTH HISTORY



Ambulatory Clinical Functionality Areas

- Organizing patient data – demographics, clinical documentation and notes, medical history
- Compiling lists – problems, medication, allergies, adverse reactions
- Receiving and displaying information – test results, consents, authorizations, clinical documents from outside the practice
- Creating orders – ordering medication or diagnostic tests; managing order sets, orders, referrals;
- Generating and recording patient-specific instructions
- Supporting decisions – presenting alerts and reminders for disease management, preventive services, wellness; checking for drug interactions and guiding appropriate responses; supporting standard care plans, guidelines and protocols; updating decision support guidelines
- Authorized sharing – managing practitioner/patient relations, enforcing confidentiality, enabling concurrent use among multiple practitioners and healthcare personnel
- Managing workflow – assigning and routing clinical tasks, managing the taking of medication and immunizations, communicating with a pharmacy
- Administrative and billing support – assist with financial and administrative coding; verifying eligibility and determining insurance coverage

Ambulatory Interoperability Areas

- **Laboratory results**
 - Receive and store lab results
 - Differentiate between a preliminary and final result
 - Process corrected results, and include information on test accuracy
 - View x-ray and other images
- **Electronic prescribing**
 - Send a new prescription
 - Approve a refill
 - Check that a medication is on the approved formulary
 - Check patient eligibility
 - Obtain medication history from the pharmacy
- **Exchange summary documents**
 - Receive and display patient record from an outside system
 - Send a patient record to an external system

Ambulatory Security Areas

- Authentication of users (proving identity)
- Controlling access based on the user role or the context of a care situation.
- Auditing access and use of records
- Encryption of any data sent out of the system.
- Protection against viruses and other malware
- Backup of data to prevent loss in case of computer failure or disaster

HL7 EHR-S Functional Model Release 2

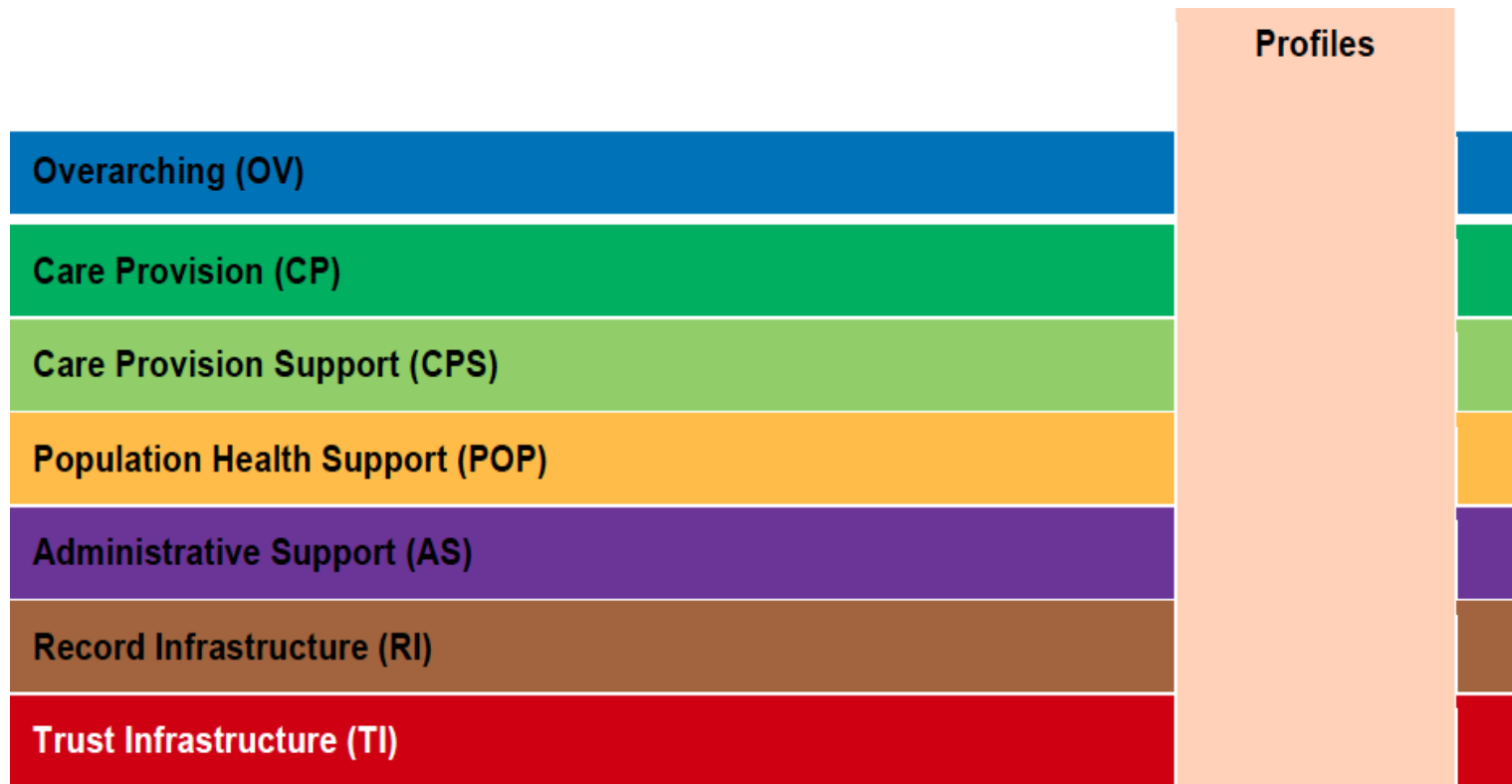


Figure 2. Profiling from the EHR-S FM.

Source: HL7 EHR System Functional Model Release 2, Copyright © 2014 Health Level Seven International

Care Provision

- Focus on functions required to provide care to a specific patient and enable hands-on delivery of healthcare
- Organized in general order of an encounter

Care Provision	CP.1	Manage Clinical History	Example child functions:	
	CP.2	Render Externally-sourced Information		
	CP.3	Manage Clinical Documentation		
	CP.4	Manage Orders		
	CP.5	Manage Results	CP.1	Manage Clinical History
	CP.6	Manage Treatment Administration	CP.1.1	Manage Patient History
	CP.7	Manage Future Care		Manage Allergy, Intolerance and Adverse Reaction List
	CP.8	Manage Patient Education & Communication	CP.1.2	Manage Medication List
	CP.9	Manage Care Coordination & Reporting	CP.1.3	Manage Problem List
			CP.1.4	Manage Strengths List
			CP.1.5	Manage Immunization List
				Manage Medical Equipment, Prosthetic/Orthotic, Device List
			CP.1.7	Manage Patient and Family Preferences
			CP.1.8	

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Types of Clinical Decision Support

Documentation

Forms/templates

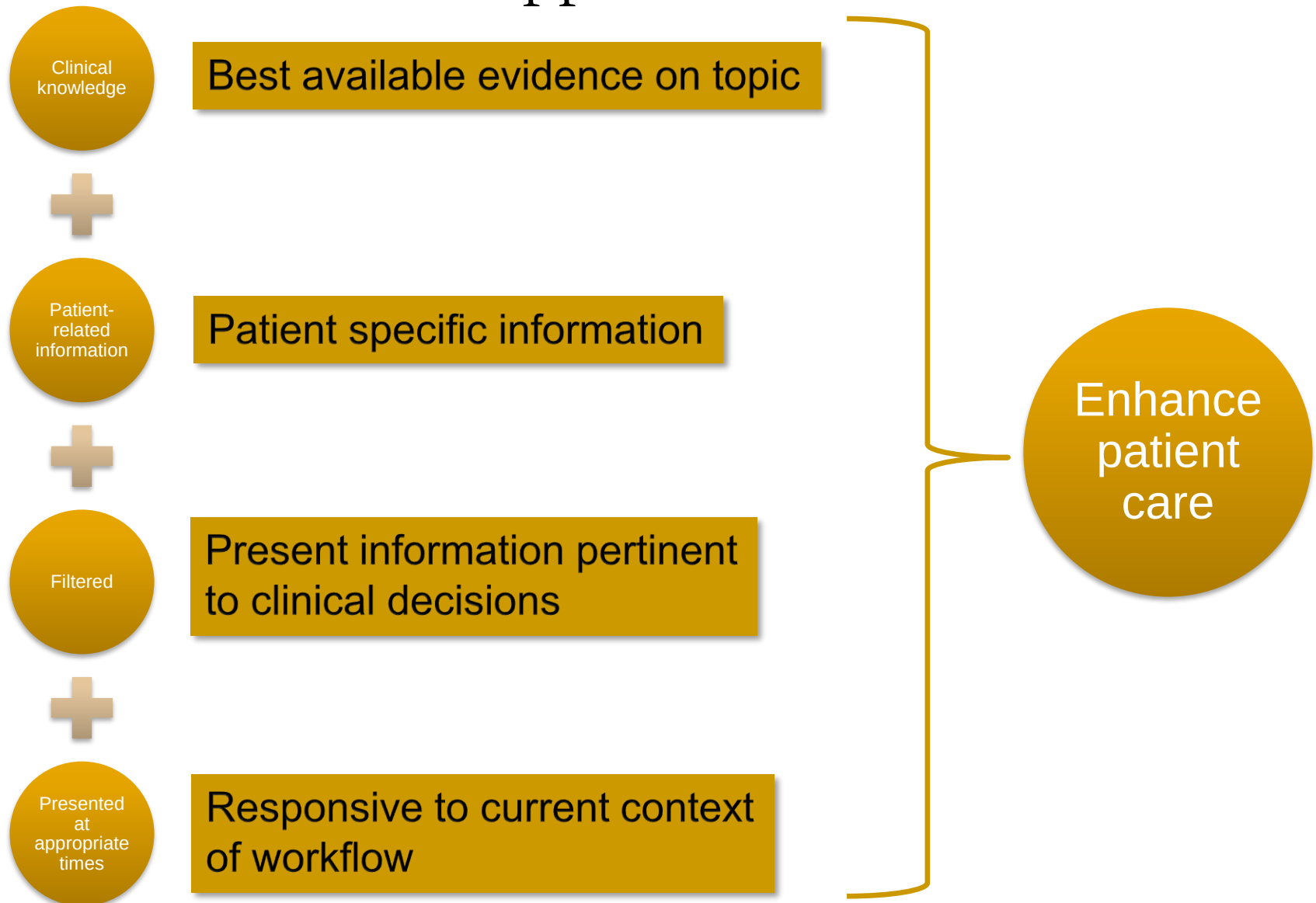
Order creation facilitators

Time-based checking and protocol/pathway support

Reactive alerts and reminders

Artificial Intelligence/Machine Learning/Natural Language Processing

Clinical Decision Support – What is it?



Using an EHR

Online use of an EHR
dr chrono



Homework – Develop an EHR Functional Requirements Worksheet

Identifying required functions is critical to implementing an EHR. Therefore, develop a high-level EHR Functional Requirements document for Virginia Women's Center's (VWC) deployment of an EHR system.

- Review the functional requirements for an EHR in the VWC case study.
- Review the CMS Medicare MIPS Quality Payment Program 2023 and now available 2024 Promoting Interoperability (MIPS/QPP/PI) requirements at:
 - <https://qpp.cms.gov/mips/promoting-interoperability>
 - <https://www.cms.gov/medicare/regulations-guidance/promoting-interoperability-programs/calendar-year-2024-program-requirements>
- Use the Microsoft Excel Functional Requirements spreadsheet on the class site (see example below). Based on the VWC case study and the CMS PI requirements, what are the top 10-12 functions needed to meet VWC's needs and the PI requirements? Please be sure to reference the importance of the VWC's needs.
- For each function, fill in the worksheet and answer:
 - How important is each function you selected to the VWC using the 1-4 scale in the spreadsheet?
 - In the Notes column, explain why you choose the 1-4 rating scale for each function and how it relates to the VWC.
- Submit the homework by submitting the completed Excel spreadsheet on the class Blackboard site. You can also copy the Excel spreadsheet into Google Sheets and submit that.
- Note: You should not work on the vendor score column for this assignment.
- To help us please use the following naming convention for your submitted file: lastname_firstname Assignment 2 and please include your name in the Excel or Google worksheet.

Homework – Develop an EHR Functional Requirements Worksheet

<i>EHR System Functional Requirements Worksheet</i>			
	Importance		
	4	Essential - Unacceptable without this function	
	3	Important - Critically important function. Not having it would be a major negative	
	2	Nice - Good to have functionality but could do without it if other functionality is good	
	1	Optional - Can do without with limited or no impact	
	Score		
	4	Fully fulfills this function with flexibility and great fit to needs	
	3	Fulfills this function well but with less optional capabilities or flexibility	
	2	Fulfills the function in a basic way	
	1	Only provides part of this function	
	0	Product does not have any of this capability	
Function	Importance	Vendor Score	Notes

Functional Requirements Grading Criteria

Learning Objective	Points
• Understanding of what the clinical part of an HIS, an EHR, does and importance of the EHR functional requirements • Identification of key functions as defined in the lecture and textbook. • Relationship of the functions to VWC	45 Points
• Importance of the functions • Description of why they are important • Relationship of the functions to achieving government goals like the MIPS, QPP, and PI (Promoting Interoperability)	45 Points
Clarity and quality of the material, use of AI documented	10 points
Total	100 Points